

BIO5105 – Short Questions (Solved) Unit-wise)

Unit 1: Cell Structure & Function

Q1. Define a cell.

Answer: A cell is the **basic structural and functional unit of life**. It can perform all processes of life independently.

Q2. Name the two main types of cells.

Answer:

1. **Prokaryotic cells** – no nucleus, DNA circular, e.g., bacteria
2. **Eukaryotic cells** – true nucleus, DNA linear, e.g., plant & animal cells

Q3. Functions of plasma membrane.

Answer:

- Acts as a **selective barrier**
- Maintains **homeostasis**
- Facilitates **communication & signaling**
- Provides **structural support**

Q4. Differences between rough and smooth ER.

Answer:

Feature	Rough ER	Smooth ER
Ribosomes Present		Absent
Function	Protein synthesis	Lipid synthesis & detox

Q5. Function of Golgi apparatus.

Answer: Modifies, sorts, and packages proteins and lipids for **secretion or delivery to organelles**.

Q6. Role of lysosomes.

Answer: Contain **digestive enzymes** to degrade macromolecules, worn-out organelles, and pathogens.

Q7. Function of mitochondria.

Answer: **ATP production** via oxidative phosphorylation; regulates **cell metabolism and apoptosis**.

Unit 2: Cell Cycle, Mitosis & Meiosis

Q1. Define cell cycle.

Answer: Series of **ordered events** through which a cell grows, replicates DNA, and divides into daughter cells.

Q2. Phases of cell cycle.

Answer:

1. **G1 phase** – cell growth
2. **S phase** – DNA replication
3. **G2 phase** – preparation for mitosis
4. **M phase** – mitosis & cytokinesis

Q3. Difference between mitosis and meiosis.

Answer:

Feature	Mitosis	Meiosis
Number of divisions	1	2
Chromosome number	Same as parent	Half of parent (haploid)
Function	Somatic cell division	Gamete formation

Q4. Checkpoints in cell cycle.

Answer:

- **G1/S checkpoint** – DNA integrity before replication
- **G2/M checkpoint** – DNA replication complete & undamaged
- **Spindle checkpoint** – Chromosome attachment before anaphase

Q5. Role of cyclins & CDKs.

Answer: Cyclins bind **CDKs** to regulate **cell cycle progression**.

Unit 3: DNA, RNA & Protein Synthesis

Q1. Define DNA.

Answer: DNA is a **double-stranded nucleic acid** carrying **genetic information**.

Q2. Types of RNA & functions.

Answer:

- **mRNA** – carries genetic code to ribosomes
- **tRNA** – carries amino acids
- **rRNA** – forms structural & catalytic part of ribosomes

Q3. Steps of DNA replication.

Answer:

1. **Unwinding** by helicase
2. **Primer synthesis** by primase

3. **Elongation** by DNA polymerase
4. **Joining Okazaki fragments** by ligase
5. **Proofreading**

Q4. Central dogma of molecular biology.

Answer: DNA → RNA → Protein

Q5. Steps of transcription.

Answer:

1. Initiation – RNA polymerase binds promoter
2. Elongation – RNA synthesized complementary to DNA
3. Termination – RNA polymerase releases transcript

Q6. Steps of translation.

Answer:

1. Initiation – ribosome assembles on mRNA
2. Elongation – amino acids added via tRNA
3. Termination – stop codon recognized, polypeptide released

Q7. Post-translational modifications.

Answer: Phosphorylation, glycosylation, acetylation – for **protein activation & targeting**

Unit 4: Cellular Organelles & Their Functions

Q1. Function of nucleus.

Answer: Stores **genetic material (DNA)**, controls **cell activities**, and site of **RNA synthesis**.

Q2. Structure & function of nucleolus.

Answer:

- Dense region inside nucleus
- Synthesizes **rRNA** and assembles **ribosome subunits**

Q3. Function of ribosomes.

Answer: Site of **protein synthesis**; can be free in cytoplasm or attached to ER.

Q4. Function of peroxisomes.

Answer:

- Detoxify harmful substances (e.g., hydrogen peroxide)
- Oxidize fatty acids

Q5. Functions of cytoskeleton.

Answer:

- Maintains **cell shape**
- Enables **motility** (cilia & flagella)
- Organizes **intracellular transport**

Q6. Microtubules, microfilaments, intermediate filaments – functions.

Answer:

- Microtubules – cell shape, chromosome separation
- Microfilaments – cell motility, endocytosis
- Intermediate filaments – mechanical strength

Q7. Function of extracellular matrix (ECM).

Answer: Provides **structural support**, mediates **cell signaling**, and influences **cell adhesion & migration**.

Unit 5: Cellular Metabolism

Q1. Define metabolism.

Answer: All **chemical reactions** in cells that **maintain life**; includes **catabolism** (breakdown) and **anabolism** (synthesis).

Q2. Define glycolysis.

Answer: Anaerobic pathway in **cytoplasm** converting **glucose** → **pyruvate**, producing **2 ATP + 2 NADH**.

Q3. Citric acid cycle (Krebs cycle) – location & function.

Answer:

- Mitochondrial matrix
- Oxidizes acetyl-CoA to **CO₂**, producing **NADH, FADH₂, ATP**

Q4. Electron Transport Chain (ETC) – location & function.

Answer:

- Inner mitochondrial membrane
- Transfers electrons to oxygen → produces **ATP & water**

Q5. Difference between aerobic & anaerobic respiration.

Answer:

Feature	Aerobic	Anaerobic
Oxygen	Required	Not required
ATP yield	~36–38 ATP	2 ATP
End product	CO ₂ + H ₂ O	Lactate or ethanol

Q6. Function of pentose phosphate pathway.

Answer: Produces **NADPH** (for biosynthesis) and **ribose-5-phosphate** (for nucleotide synthesis).

Q7. Beta-oxidation of fatty acids – location & purpose.

Answer:

- Mitochondria
- Converts fatty acids → **acetyl-CoA** for Krebs cycle

Q8. Urea cycle – location & function.

Answer:

- Liver
- Detoxifies **ammonia** → urea for excretion

Q9. Gluconeogenesis – definition.

Answer: Synthesis of **glucose from non-carbohydrate sources** (e.g., lactate, glycerol, amino acids).

Q10. Difference between glucogenic & ketogenic amino acids.

Answer:

- Glucogenic → converted to glucose
- Ketogenic → converted to ketone bodies

Unit 6: Enzymes & Their Regulation

Q1. Define enzyme.

Answer: Biological **catalyst** that speeds up chemical reactions without being consumed.

Q2. Factors affecting enzyme activity.

Answer:

- Temperature
- pH
- Substrate concentration
- Presence of inhibitors or activators

Q3. Define V_{max} & K_m.

Answer:

- **V_{max}** – maximum reaction rate at saturating substrate
- **K_m** – substrate concentration at ½ **V_{max}**, indicates enzyme affinity

Q4. Allosteric enzymes – definition & regulation.

Answer: Enzymes with **regulatory sites** where effectors bind → **activate or inhibit** enzyme function.

Q5. Feedback inhibition – definition.

Answer: End product of a metabolic pathway inhibits **first enzyme**, preventing overproduction.

Q6. Cofactors & coenzymes – difference.

Answer:

- **Cofactors** – inorganic ions (Mg^{2+} , Zn^{2+})
- **Coenzymes** – organic molecules (NAD^+ , FAD, CoA)

Q7. Function of kinases & phosphatases.

Answer:

- **Kinases** – add phosphate groups to proteins
 - **Phosphatases** – remove phosphate groups
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Unit 7: DNA Repair & Recombination

Q1. Define DNA repair.

Answer: Cellular processes that **detect and correct DNA damage** to maintain **genome integrity**.

Q2. Types of DNA damage.

Answer:

- Single-strand breaks
- Double-strand breaks
- Base modifications
- UV-induced pyrimidine dimers

Q3. Nucleotide Excision Repair (NER) – function.

Answer: Removes **bulky DNA lesions** (e.g., UV-induced thymine dimers) and fills gap with **DNA polymerase**.

Q4. Base Excision Repair (BER) – function.

Answer: Corrects **single-base damage** using **glycosylases**, AP endonuclease, and DNA polymerase.

Q5. Mismatch Repair (MMR) – function.

Answer: Fixes **replication errors** like mispaired bases.

Q6. Homologous recombination – function.

Answer: Exchange of **DNA segments between homologous chromosomes** to repair double-strand breaks or generate genetic diversity.

Q7. Non-Homologous End Joining (NHEJ) – function.

Answer: Repairs **double-strand breaks** without requiring a homologous template.

Unit 8: Protein Synthesis & Post-Translational Modifications

Q1. Define translation.

Answer: Process of **synthesizing proteins** from **mRNA template** using **ribosomes** and **tRNA**.

Q2. Steps of translation.

Answer:

1. Initiation – ribosome assembles at start codon
2. Elongation – amino acids added to growing polypeptide
3. Termination – release of polypeptide at stop codon

Q3. Post-translational modifications – examples & purpose.

Answer:

- Phosphorylation – regulates activity
- Glycosylation – targets proteins to membrane or secretion
- Acetylation – affects stability & DNA binding
- Ubiquitination – targets proteins for degradation

Q4. Role of chaperones.

Answer: Assist in **protein folding** and prevent misfolded proteins from aggregation.

Q5. ER-associated degradation (ERAD) – function.

Answer: Removes **misfolded proteins from ER** and targets them for degradation.

Q6. Secretory pathway – steps.

Answer: ER → Golgi → vesicles → **plasma membrane** for protein/lipid secretion.

Q7. Clathrin-coated vesicles – function.

Answer: Mediate **receptor-mediated endocytosis** for specific molecules.

Unit 9: Cell Signaling & Communication

Q1. Define cell signaling.

Answer: Process by which cells **detect and respond to signals** from their environment or other cells.

Q2. Types of signaling.

Answer:

- **Autocrine** – cell signals itself
- **Paracrine** – signals neighboring cells
- **Endocrine** – signals distant cells via bloodstream
- **Juxtacrine** – direct contact between cells

Q3. Receptors – types & function.

Answer:

- **G-protein coupled receptors (GPCRs)** – activate second messengers (cAMP, Ca^{2+})
- **Receptor tyrosine kinases (RTKs)** – phosphorylate proteins → signal growth/division
- **Ion channels** – respond to voltage or ligands

Q4. Second messengers – examples & function.

Answer:

- **cAMP, cGMP, Ca^{2+}**
- Amplify signal inside cell to activate downstream responses

Q5. Steroid vs peptide hormone signaling.

Answer:

- Steroid – intracellular receptor → gene transcription
- Peptide – membrane receptor → second messengers

Q6. Gap junctions – function.

Answer: Direct **cytoplasmic communication** between neighboring cells.

Q7. Tight junctions – function.

Answer: Seal adjacent cells to **prevent leakage** between compartments.

Q8. Desmosomes – function.

Answer: Provide **mechanical strength** to tissues by connecting intermediate filaments.

Unit 10: Mitochondria & Energy Metabolism

Q1. Function of mitochondria.

Answer: Produce **ATP** via **oxidative phosphorylation**; regulate **apoptosis** and metabolic pathways.

Q2. Mitochondrial structure – key components.

Answer:

- **Outer membrane** – permeable barrier
- **Inner membrane** – contains ETC and ATP synthase
- **Matrix** – contains Krebs cycle enzymes, mtDNA, ribosomes

Q3. Mitochondrial DNA – features.

Answer: Circular, maternally inherited, encodes some **proteins, tRNAs, rRNAs**.

Q4. Electron transport chain – function.

Answer: Transfers electrons from NADH/FADH₂ to oxygen → generates **proton gradient** → **ATP synthesis**.

Q5. ATP synthase – function.

Answer: Uses proton gradient to **synthesize ATP** from ADP + Pi.

Q6. Reactive oxygen species (ROS) – source & detoxification.

Answer:

- Source: ETC leakage in mitochondria
 - Detoxification: **Catalase, superoxide dismutase, glutathione peroxidase**
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Unit 11: Chloroplasts & Photosynthesis

Q1. Function of chloroplast.

Answer: Conduct **photosynthesis**; convert light energy → chemical energy (glucose).

Q2. Chloroplast structure – key components.

Answer:

- **Thylakoids** – light-dependent reactions
- **Stroma** – Calvin cycle
- **Grana** – stacks of thylakoids

Q3. Chloroplast DNA – inheritance.

Answer: Mostly **maternal**, circular DNA encoding some photosynthetic proteins.

Q4. Photosynthesis – two main stages.

Answer:

1. **Light reactions** – produce ATP & NADPH (thylakoid)
2. **Calvin cycle** – fixes CO₂ into glucose (stroma)

Q5. C4 vs CAM plants – difference.

Answer:

Feature	C4	CAM
CO ₂ fixation	Mesophyll & bundle sheath	Night (fix), Day (Calvin)
Advantage	Reduce photorespiration	Conserve water

Unit 12: Cellular Signaling & Regulation of Metabolism

Q1. Define second messengers.

Answer: Small molecules that **transmit signals** from receptors → target molecules inside cell (e.g., cAMP, Ca²⁺).

Q2. Key regulators of metabolism.

Answer:

- **Allosteric enzymes** – activity regulated by substrates/products
- **Hormones** – insulin, glucagon, epinephrine
- **Cofactors & coenzymes** – NAD^+ , FAD, ATP

Q3. Feedback inhibition – definition & example.

Answer: End product of pathway inhibits **first enzyme** to prevent overproduction. Example: **ATP inhibits phosphofructokinase in glycolysis.**

Q4. Role of kinases & phosphatases in regulation.

Answer:

- Kinases – **phosphorylate proteins** to activate/inhibit enzymes
- Phosphatases – **remove phosphate groups**

Q5. Signal transduction – general steps.

Answer:

1. Ligand binds receptor
2. Receptor activates **second messenger**
3. Signal amplified
4. Target proteins activated → cellular response

Q6. Hormonal regulation – steroid vs peptide hormones.

Answer:

- Steroid → intracellular receptor → gene transcription
- Peptide → membrane receptor → second messenger → enzyme activation

Unit 13: Enzyme Kinetics & Regulation

Q1. Define enzyme kinetics.

Answer: Study of the **rate of enzymatic reactions** and factors affecting them.

Q2. Michaelis-Menten equation – definition.

Answer: Describes the **relationship between reaction rate (v), substrate concentration ([S]), V_{max} , and K_m :**

$$v = \frac{V_{\text{max}}[S]}{K_m + [S]}$$

Q3. Define V_{max} & K_m .

Answer:

- **V_{max}** – maximum reaction velocity

- **K_m** – substrate concentration at ½ **V_{max}**, indicates enzyme affinity

Q4. Types of enzyme inhibition.

Answer:

- **Competitive** – inhibitor binds active site, increases K_m
- **Non-competitive** – binds allosteric site, decreases V_{max}
- **Uncompetitive** – binds ES complex, decreases V_{max} & K_m

Q5. Allosteric regulation – definition & example.

Answer: Enzyme activity regulated by **binding of effectors at sites other than active site.**

Example: **ATP inhibits phosphofructokinase**

Q6. Cofactors vs coenzymes – difference.

Answer:

- **Cofactors** – inorganic ions (Mg²⁺ , Zn²⁺)
- **Coenzymes** – organic molecules (NAD⁺ , FAD, CoA)

Q7. Feedback inhibition – definition & example.

Answer: End product inhibits **first enzyme** of pathway. Example: **ATP inhibits phosphofructokinase in glycolysis**

Unit 14: Nitrogen Metabolism & Amino Acid Catabolism

Q1. Define nitrogen balance.

Answer: Difference between **nitrogen intake & output.**

- Positive → growth/repair
- Negative → starvation/illness

Q2. Essential vs non-essential amino acids.

Answer:

- **Essential** – must be obtained from diet
- **Non-essential** – synthesized by body

Q3. Transamination – definition & role.

Answer: Transfer of **amino group** from amino acid → keto acid. Critical for **amino acid catabolism.**

Q4. Oxidative deamination – definition & role.

Answer: Removal of **amino group as ammonia** (via glutamate dehydrogenase).

Q5. Glucogenic vs ketogenic amino acids.

Answer:

- Glucogenic → converted to glucose

- Ketogenic → converted to acetyl-CoA → ketone bodies

Q6. Urea cycle – location & function.

Answer:

- Liver mitochondria & cytoplasm
- Detoxifies **ammonia** → **urea**

Q7. Clinical significance – high ammonia.

Answer: Hyperammonemia → neurological toxicity; caused by **urea cycle defects**

Unit 15: Biosynthesis Pathways

Q1. Fatty acid synthesis – location & key steps.

Answer:

- Cytoplasm
- Acetyl-CoA → Malonyl-CoA → chain elongation
- Requires **NADPH**

Q2. Fatty acid oxidation – location & purpose.

Answer:

- Mitochondria
- Breaks fatty acids → **acetyl-CoA** for ATP production

Q3. Cholesterol biosynthesis – key enzyme.

Answer: HMG-CoA reductase

Q4. Ketone body formation – definition & location.

Answer: Liver mitochondria convert **acetyl-CoA** → **ketone bodies** during fasting.

Q5. Gluconeogenesis – definition & precursors.

Answer:

- Synthesis of glucose from non-carbohydrate sources
- Precursors: lactate, glycerol, glucogenic amino acids

Q6. Pentose phosphate pathway – main purpose.

Answer: Produce **NADPH** (for biosynthesis) and **ribose-5-phosphate** (for nucleotides).

Q7. Key regulators of metabolism – examples.

Answer:

- Hormones: insulin, glucagon, epinephrine
- Allosteric enzymes: phosphofructokinase, pyruvate kinase
- Energy charge: ATP/ADP/NADH ratios

Unit 16: Signal Transduction & Cellular Communication

Q1. Define signal transduction.

Answer: Process by which **cells detect signals** from their environment and **respond** via intracellular pathways.

Q2. Types of cell signaling.

Answer:

- **Autocrine** – acts on the same cell
- **Paracrine** – acts on neighboring cells
- **Endocrine** – acts on distant cells via bloodstream
- **Juxtacrine** – requires direct cell-to-cell contact

Q3. Receptor types – examples & functions.

Answer:

- **G-protein coupled receptors (GPCRs)** – activate second messengers (cAMP, Ca^{2+})
- **Receptor tyrosine kinases (RTKs)** – phosphorylate proteins → regulate growth/division
- **Ion channels** – respond to voltage or ligand

Q4. Second messengers – definition & examples.

Answer: Small molecules that **amplify and transmit signals**: cAMP, cGMP, Ca^{2+}

Q5. Hormone signaling – steroid vs peptide.

Answer:

- Steroid → intracellular receptor → gene transcription
- Peptide → membrane receptor → second messenger → enzyme activation

Q6. Cell junctions – types & function.

Answer:

- **Gap junctions** – cytoplasmic communication
- **Tight junctions** – seal cells to prevent leakage
- **Desmosomes** – provide mechanical strength

Unit 17: Mitochondrial & Chloroplast Dynamics

Q1. Mitochondrial dynamics – define fusion & fission.

Answer:

- **Fusion** – mitochondria merge → network formation, maintain function
- **Fission** – mitochondria divide → remove damaged parts, facilitate mitosis

Q2. Mitophagy – definition & purpose.

Answer: Selective degradation of damaged mitochondria via **autophagy**

Q3. Chloroplast inheritance – maternal or paternal?

Answer: Mostly **maternal**, circular DNA encodes photosynthetic proteins

Q4. Chloroplast – light vs dark reactions.

Answer:

- Light reactions (thylakoid) → ATP & NADPH
- Calvin cycle (stroma) → CO₂ fixation → glucose

Q5. C4 vs CAM photosynthesis – advantage.

Answer:

- **C4** – minimizes photorespiration in hot climates
- **CAM** – conserves water in arid conditions

Q6. Reactive oxygen species (ROS) in organelles – source & detox.

Answer:

- Source: mitochondria & chloroplast ETC leakage
 - Detox: catalase, superoxide dismutase, glutathione peroxidase
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Unit 18: Integration of Metabolism & Cellular Homeostasis

Q1. Define metabolic integration.

Answer: Coordination of **catabolic & anabolic pathways** to maintain energy and biomolecule balance.

Q2. Key energy molecules – role of ATP, NADH, NADPH.

Answer:

- **ATP** – universal energy currency
- **NADH** – electrons for oxidative phosphorylation
- **NADPH** – reducing power for biosynthesis

Q3. Glucose metabolism regulation – key points.

Answer:

- **Glycolysis** → regulated by phosphofructokinase (ATP inhibits, AMP activates)
- **Gluconeogenesis** → regulated by fructose-1,6-bisphosphatase
- **Glycogen metabolism** → glycogen synthase & phosphorylase

Q4. Lipid metabolism – key regulation.

Answer:

- Fatty acid synthesis – citrate & NADPH

- β -oxidation – inhibited by malonyl-CoA
- Ketone bodies – produced during fasting

Q5. Nitrogen metabolism – homeostasis.

Answer:

- Ammonia detoxified via **urea cycle** in liver
- Positive nitrogen balance $\rightarrow$ growth
- Negative nitrogen balance $\rightarrow$ starvation/illness

Q6. Feedback regulation – examples.

Answer:

- ATP inhibits phosphofructokinase
 - NADPH inhibits glucose-6-phosphate dehydrogenase
-